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Hyper-Local Surplus Food Rescue & Safety-Compliant Redistribution Network

🎯 UN SDG 2: Zero Hunger | 🌱 UN SDG 12: Responsible Consumption | ⚡ 15–30 Min Rescue Velocity

01
TARGET ECOSYSTEM

Focused Campus Pilot:
- UIU Central Cafeteria
- Dhanmondi & Badda Local Caterers
- Nearby Orphanages & Day Shelters

02
CORE SOFTWARE GOAL

Eliminate Cooked Food Dumping:
- Sub-60s rapid listing module
- 3–5 km proximity radar GIS filter
- 4-Digit OTP handshake delivery verification

03
ACADEMIC SCOPE

16-Week Implementation:
- Next.js 16 SSR & Prisma ORM
- PostgreSQL ACID claim state engine
- Leaflet.js GIS without paid API billing

1. Problem Identification: What Problem Are We Solving?

The critical paradox of urban food waste vs. street hunger in Dhaka City

CORE DIAGNOSIS

Why Does Good Food Go to Waste?

In Dhaka, food waste is **NOT** caused by food shortage.

It is fundamentally an **INFORMATION, COORDINATION, and MICRO-LOGISTICS failure**.

Institutional kitchens cook in bulk. When trays remain at night, donors have zero rapid, verified channels to reach hungry communities within the critical biological safety window.

01 THE DAILY MIDNIGHT FOOD DUMP

University cafeterias (e.g. UIU Central Cafeteria), event caterers, and eateries discard 40–80 kg of untouched hot meals every night into municipal dumpsters due to zero instant charity links.

02 THE 4-HOUR BIOLOGICAL SAFETY CLOCK

Cooked food enters the bacterial danger zone (5°C–60°C). Under Bangladesh Food Safety Authority (BFSA) guidelines, cooked meals must be eaten within 4 hours. Manual coordination takes 5+ hours.

03 THE MISSING MICRO-LOGISTICS LINK

Nearby orphanages and shelters lack vehicles or night staff. Food is available just 2 km away, but there is zero automated courier dispatch system to transport it in time.

💡 **KEY TAKEAWAY: Food surplus and severe hunger exist within 3 km of each other, but lack an automated software platform to bridge the 4-hour safety window.**

1. Evidence & Data: Backed by Hard Statistics

Authoritative research on food waste, biological danger limits, and urban food insecurity

10.7M Tons
ANNUAL FOOD WASTE
UNEP Food Waste Index 2024 (BD)

4 Hours
BIOLOGICAL SAFETY LIMIT
BFSA Danger Zone (5°C – 60°C)

3.8 Million
URBAN FOOD-INSECURE
BBS & World Bank Slum Report

0% Digital
CAMPUS TRACEABILITY TODAY
Zero Audit Trail or Safety Records

01

SOCIAL IMPACT: ZERO HUNGER (SDG 2)

Directly feeds destitute children, orphanages, and street laborers with preserved human dignity. Rescuing a single 40 kg batch feeds **80–100 hungry people** every night.

02

HEALTH & SAFETY: BFSA COMPLIANCE

Enforces mandatory digital timestamping and temperature monitoring. Food exceeding the 4-hour threshold is automatically locked out, completely eliminating food poisoning risks.

03

ENVIRONMENT: METHANE PREVENTION (SDG 12)

Diverts tons of decomposing organic food from open city landfills in Aminbazar and Matuail, slashing lethal greenhouse gas emissions and urban environmental decay.

Data Sources: UNEP Food Waste Index 2024, Bangladesh Food Safety Authority (BFSA), BBS Urban Survey

2. Solution: The Core Idea & 3-Sided Ecosystem

A synchronized hyper-local network engineered for 15-30 minute rescue turnaround

☀ ONE SINGLE LINE (EXPLAINING THE PROJECT TO A KID OR PARENT):

"An Uber-like platform where university cafeterias post extra hot food, and student volunteers bike it to nearby orphanages in 30 minutes before it spoils!"

01

FOOD DONORS

Cafeterias, Caterers & Restaurants

- **60-Second Posting:** Quick 3-field listing form.
- **AI Spoilage Gauge:** Visual countdown dial.
- **Meal Converter:** '40 kg Biryani = 80 meals'.

02

VERIFIED NGOS

Orphanages & Local Shelters

- **Proximity Radar Map:** Filtered to 3-5 km.
- **1-Click Claim Engine:** Instant reservation.
- **Digital CSR Receipts:** Automated audit reports.

03

STUDENT VOLUNTEERS

Campus Cyclists & Bikers Squad

- **Uber-Style Push Alerts:** Notified during commutes.
- **15-30 Min Velocity:** Rapid drop-off.
- **Thermal Bag Check:** Insulated food safety.

2. Main Features: What Will the End System Perform?

Four engineered modules eliminating the primary failure points in food rescue

01 AI SPOILAGE METER & TIMEOUT LOCK

Input: Cooking timestamp & storage type.
Mechanism: Dynamic biological decay countdown enforcing the BFSA 4-hour window.
Output: Real-time visual dial; automatically de-lists batches when time runs out to ensure 100% food safety.

02 3-5 KM HYPER-LOCAL RADAR MAP

Input: GPS coordinates of donors and recipient shelters.
Mechanism: Client-side Haversine circular radius queries on OpenStreetMap.
Output: Eliminates Dhaka traffic gridlock by strictly matching donations within a 15-minute cycling/walking radius.

03 UBER-STYLE VOLUNTEER DISPATCH

Input: Active verified claims & volunteer locations.
Mechanism: Instant broadcast push alerts to registered student cyclists/bikers.
Output: Provides turn-by-turn route directions, donor contact, pickup deadline, and thermal bag checklist.

04 4-DIGIT OTP & QR ANTI-FRAUD PROTOCOL

Input: Donor secret PIN & NGO receipt token.
Mechanism: Dual-layer cryptographic handshake at pickup and drop-off.
Output: Eliminates food diversion or black-market resale; guarantees food reaches intended children with 100% audit trail.

WORKFLOW: 60s Post ➔ AI Spoilage Verification ➔ 3km Radar Alert ➔ Student Biker Dispatch ➔ OTP Handshake Handover

3. Plan of Action: High-Level Technology Overview

Modern open-source stack chosen for rapid response time, relational integrity, and zero operating costs

SYSTEM ARCHITECTURE & DATA FLOW PIPELINE



01

FRONTEND & MOBILE PWA

Next.js 16 + React 19 + Tailwind CSS

Role: Unified web portal & mobile PWA for all 3 user roles.

Why Chosen:
- Sub-100ms cold start times over 4G cellular data
- Progressive Web App (PWA) eliminates app-store install barrier
- Tailwind CSS v4 provides lightweight, responsive UI

02

BACKEND & DATA LAYER

Prisma ORM + PostgreSQL + TypeScript

Role: Data persistence, auth & claim state machine.

Why Chosen:
- ACID transactions prevent race conditions (no double-claims)
- End-to-end type safety eliminates runtime schema errors
- Relational integrity links Donors, NGOs, and Couriers

03

GEOSPATIAL ENGINE

Leaflet.js + OpenStreetMap (OSM)

Role: Interactive radar map, circular radius query & distance.

Why Chosen:
- 100% open-source without Google Maps billing/credit card limits
- Fast client-side Haversine circular radius queries
- Low memory footprint runs smoothly on student smartphones

4. Project Timeline: 16-Week Implementation Roadmap

Structured semester progression from system architecture to live UIU campus field pilot and defense

01

PHASE 1: ARCHITECTURE

WEEKS 1 – 4

- System Requirements (SRS)
- Prisma relational schema models
- 3-Portal responsive UI setup
- Role-based authentication

Status: 100% Completed

02

PHASE 2: CORE LOGIC

WEEKS 5 – 8

- 60s listing form & meal calculator
- AI Spoilage dynamic countdown
- Leaflet geospatial radar (3km)
- 4-Digit OTP & QR handshake

Status: 90% Completed

03

PHASE 3: CAMPUS PILOT

WEEKS 9 – 12

- UIU Cafeteria live trial run
- Student cyclist squad onboarding
- Satarkul orphanage delivery test
- Turnaround latency tuning

Target: Live Field Run

04

PHASE 4: AUDIT & DEFENSE

WEEKS 13 – 16

- Concurrency & stress testing
- Anti-fraud security penetration test
- Project impact analytics report
- Final SE Lab defense presentation

Target: Final Defense

🚀 **Ready to Transform Campus Surplus into Immediate Hunger Relief**

Zero commercial barrier • Real community impact at UIU • Rigorous Software Engineering execution

Q & A SESSION

Open for Evaluation, Faculty Guidance & Feedback